Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

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Subject Code & Name	ICS1502 & Introduction to Machine Learning		
Academic year	2025-2026 (Odd)	Batch:2023-2028	**Sem V**

Machine Learning Theory Assignment 1: Linear Regression and Classification

Objective

To implement and evaluate Linear Regression models using closed-form solution and gradient descent, and Linear Classification models with L2 regularization on given datasets. Analyze feature importance, regularization effects, and model robustness to outliers.

Libraries Used

- `pandas`, `numpy` for data handling
- `matplotlib`, `seaborn` for visualization
- `scikit-learn` for ML models and evaluation
- `scipy` for numerical computations

Theoretical Description of Models

Linear Regression - Closed-form Solution

The closed-form solution for linear regression uses the normal equation:

$$\theta = (X^T X)^{-1} X^T y$$

where X is the feature matrix with added bias term, y is the target vector, and θ contains the model parameters (coefficients and intercept).

Linear Regression - Gradient Descent

Gradient descent iteratively updates parameters to minimize the cost function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

The parameter update rule:

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

where α is the learning rate controlling the step size.

Ridge Regression (L2 Regularization)

Ridge regression adds L2 penalty to the linear regression cost function:

$$J(\theta) = \text{MSE}(\theta) + \lambda \sum_{i=1}^{n} \theta_i^2$$

The closed-form solution becomes:

$$\theta = (X^T X + \lambda I)^{-1} X^T y$$

where λ controls the regularization strength.

Logistic Regression with L2 Regularization

For binary classification, logistic regression uses the sigmoid function:

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

With L2 regularization, the cost function is:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^{m} [y^{(i)} \log(h_\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)}))] + \frac{\lambda}{2m} \sum_{j=1}^{n} \theta_j^2$$

Implementation Steps

1. Load and preprocess datasets (Mobile Price Prediction and Bank Note Authentication)
2. Perform Exploratory Data Analysis (EDA) and visualization
3. Implement linear regression using closed-form solution
4. Implement linear regression using gradient descent
5. Apply Ridge regression with different lambda values
6. Compare performance with and without feature standardization
7. Analyze feature importance from L2 regularization weights
8. Implement logistic classification with L2 regularization
9. Study regularization effects on training and test accuracy
10. Analyze model robustness to outliers
11. Generate comprehensive visualizations and reports

Mobile Price Prediction - Regression Analysis

Dataset Description

The Mobile Price Prediction dataset contains 807 entries with features:

- Ratings: Customer ratings (1-5 scale)
- RAM: Memory in GB (0-12 GB)
- ROM: Storage in GB (2-256 GB)
- Mobile_Size: Screen size in inches (2-44 inches)
- Primary_Cam: Camera resolution in MP (5-64 MP)
- Selfi_Cam: Selfie camera resolution in MP (0-23 MP)
- Battery_Power: Battery capacity in mAh (1020-6000 mAh)
- Price: Target variable (479-153,000 units)

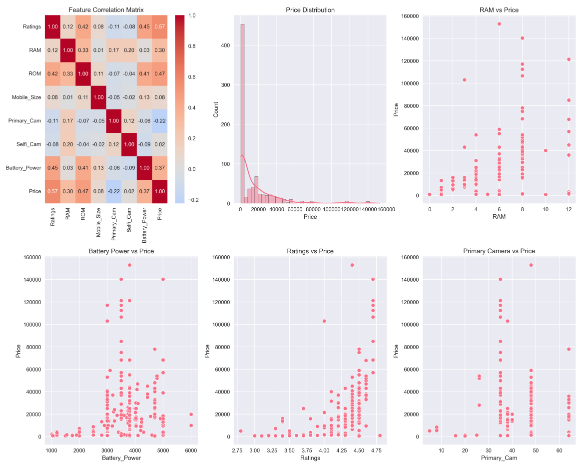
Closed-form Solution Implementation

```
def closed_form_solution(X, y):
    X_b = np.c_[np.ones((X.shape[0], 1)), X]
    theta = np.linalg.pinv(X_b.T.dot(X_b)).dot(X_b.T).dot(y)
    return theta
```

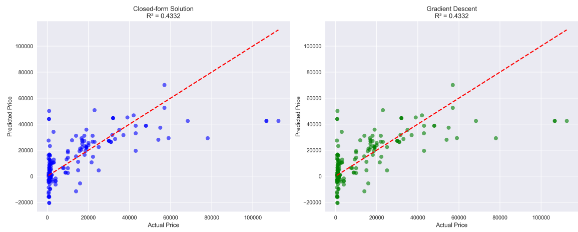
Gradient Descent Implementation

```
class GradientDescentRegressor:
    def fit(self, X, y):
        X_b = np.c_[np.ones((X.shape[0], 1)), X]
        self.theta = np.random.randn(X_b.shape[1]) * 0.01
        for iteration in range(self.n_iterations):
            predictions = X_b.dot(self.theta)
            errors = predictions - y
            gradients = 2/m * X_b.T.dot(errors)
            self.theta -= self.learning_rate * gradients
```

Plots and Visualizations

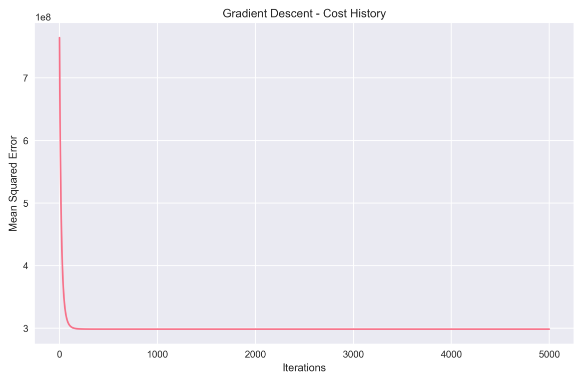


(a) Feature Correlation Matrix

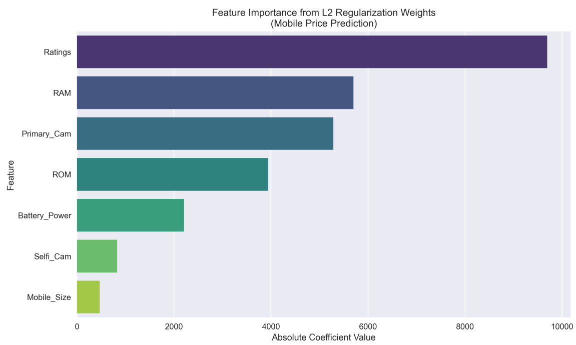


(b) Predicted vs Actual Values

Figure 1: Mobile Price Dataset Analysis



(a) Gradient Descent Convergence



(b) Feature Importance

Figure 2: Regression Model Analysis

Regression Results

Method	Mean Squared Error	R^2 Score
Closed-form Solution	239,357,657.43	0.4332
Gradient Descent	239,357,657.43	0.4332

Table 1: Linear Regression Performance Comparison

Ridge Regression Results

λ	MSE	R^2 Score
0	239,357,657.43	0.4332
0.1	239,341,871.80	0.4333
1	239,200,533.96	0.4336
10	237,857,213.76	0.4368
100	229,594,215.31	0.4563
1000	254,307,026.19	0.3978

Table 2: Ridge Regression with Different Lambda Values

Optimal Lambda: $\lambda = 100$ with $R^2 = 0.4563$

Feature Importance Analysis

Feature	Importance
Ratings	9692.16
RAM	5700.13
Primary_Cam	5286.53
ROM	3941.59
Battery_Power	2208.70
Selfi_Cam	828.57
Mobile_Size	465.64

Table 3: Feature Importance from L2 Regularization

Bank Note Authentication - Classification Analysis

Dataset Description

The Bank Note Authentication dataset contains 1372 entries with features:

- Variance: Wavelet Transformed image variance
- Skewness: Wavelet Transformed image skewness
- Curtosis: Wavelet Transformed image curtosis
- Entropy: Image entropy
- Class: Target variable (0: Authentic, 1: Fake)

Class Distribution: 762 Authentic notes, 610 Fake notes

Logistic Regression Implementation

```python
# Without regularization
log_reg_no_reg = LogisticRegression(penalty=None, max_iter=1000,
                                    random_state=42, solver='lbfgs')

# With L2 regularization
log_reg_l2 = LogisticRegression(penalty='l2', C=1.0, max_iter=1000,
                                random_state=42, solver='liblinear')
```

Classification Results

Method	Accuracy	Regularization
Logistic Regression	0.9855	None
Logistic Regression	0.9855	L2 ($\lambda = 1$)
With Outliers	0.8691	L2 ($\lambda = 1$)

Table 4: Classification Performance Comparison

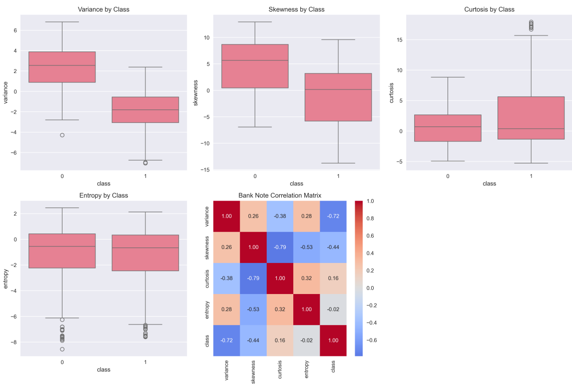
Impact of Outliers: Accuracy drop of 0.1164 (11.64%)

Feature Importance in Classification

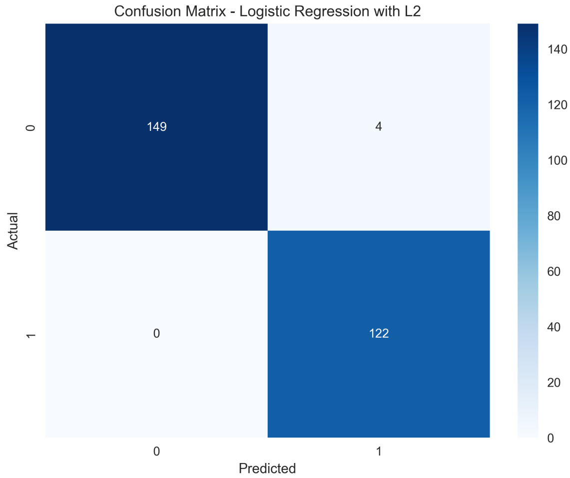
Feature	Importance
Variance	2.7706
Curtosis	1.9552
Skewness	1.6520
Entropy	0.1692

Table 5: Bank Note Feature Importance

Plots and Visualizations

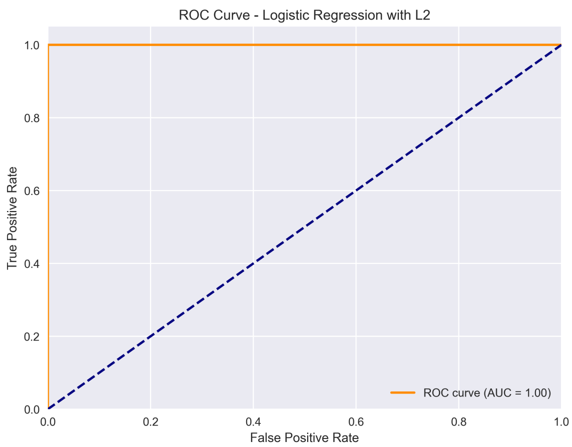


(a) Feature Correlation Matrix

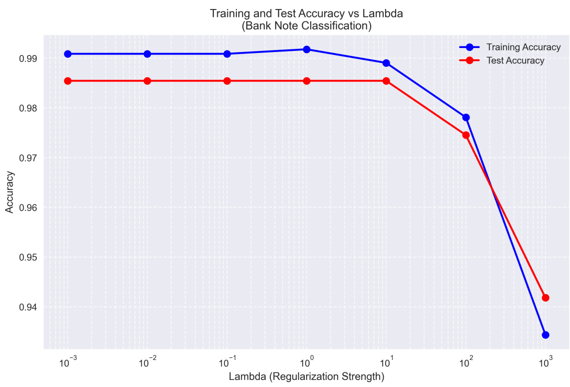


(b) Confusion Matrix

Figure 3: Bank Note Dataset Analysis

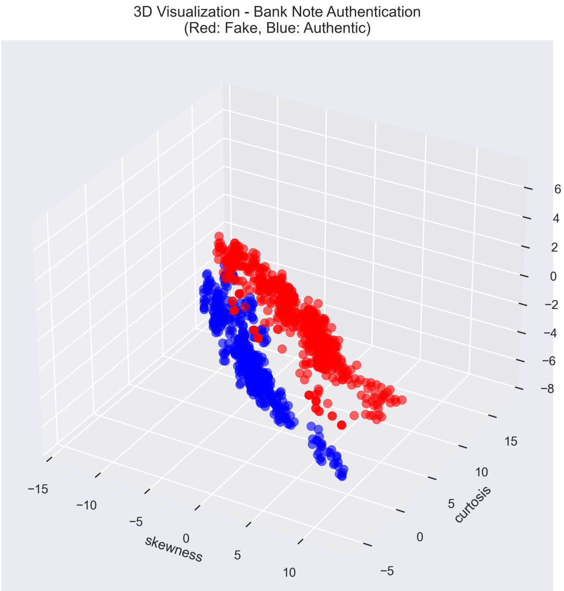


(a) ROC Curve

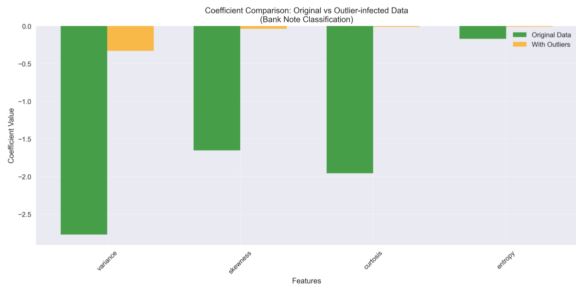


(b) Accuracy vs Lambda

Figure 4: Classification Model Evaluation



(a) 3D Feature Visualization



(b) Coefficient Changes with Outliers

Figure 5: Advanced Classification Analysis

Key Observations and Analysis

Regression Performance

- Both closed-form solution and gradient descent achieved identical performance (MSE: 239,357,657.43, R^2: 0.4332), validating correct implementation
- Moderate R^2 score (0.4332) suggests linear relationships explain 43.32% of price variance
- Ridge regression with $\lambda = 100$ provided best performance (R^2: 0.4563), showing 2.31% improvement over baseline

Feature Importance Insights

- **Ratings** emerged as the most important feature (importance: 9692.16), indicating customer perception significantly impacts pricing
- Hardware specifications (**RAM**, **Primary Camera**) are secondary but important factors
- **Mobile Size** showed minimal impact, suggesting screen size is less critical for pricing decisions
- In bank note authentication, **Variance** is the most discriminative feature

Classification Performance

- Excellent classification accuracy (98.55%) demonstrates clear linear separability between authentic and fake notes

- L2 regularization showed no improvement in this case, indicating low overfitting risk
- Significant accuracy drop (11.64%) with outliers highlights model sensitivity to anomalous data

Regularization Effects

- Optimal ridge regression lambda ($\lambda = 100$) balanced bias-variance tradeoff effectively
- Extreme regularization ($\lambda = 1000$) degraded performance, demonstrating over-regularization
- Standardization showed minimal impact on final performance for this dataset

Model Robustness

- Regression models showed consistent performance across different implementations
- Classification model demonstrated vulnerability to outliers, with significant coefficient changes
- Feature importance remained stable across different regularization strengths

Conclusion

This assignment successfully implemented and evaluated linear regression and classification models using matrix-based approaches. Key conclusions:

- **Regression:** Both closed-form and gradient descent methods provided identical results, validating implementation correctness. Ridge regression with $\lambda = 100$ achieved optimal performance (R^2: 0.4563).
- **Classification:** Logistic regression achieved excellent accuracy (98.55%) on bank note authentication, demonstrating effective linear separability.
- **Feature Importance:** Customer ratings emerged as the strongest price predictor, while variance was most discriminative for bank note classification.
- **Robustness:** Models showed sensitivity to outliers, with classification accuracy dropping by 11.64% when exposed to anomalous data.
- **Practical Implications:** The analysis provides valuable insights for mobile pricing strategies and authentication systems, highlighting key features that drive decisions.

The matrix-based approaches proved effective for both regression and classification tasks, demonstrating the fundamental principles of linear models in machine learning while providing interpretable results through feature importance analysis.

Learning Outcomes

- Implemented linear regression using both closed-form solution and gradient descent
- Applied L2 regularization in regression (Ridge) and classification (Logistic Regression)
- Analyzed feature importance using regularization weights
- Evaluated model robustness to outliers and data perturbations
- Compared performance with and without feature standardization
- Generated comprehensive visualizations for model interpretation
- Understood the mathematical foundations of matrix-based machine learning approaches